Software Requirements Specification

for

Team 17

Version 1.0 approved

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KAIST

April 8, 2026

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1. Introduction

1.1 Purpose

The objective of this document is to define the functional and non-functional requirements for a system that allows KAIST members to choose their meal easily and share menu feedback. This platform is designed to streamline the distribution of meal information and gather authentic user feedback. The system aims to improve food quality through user-driven data and enhance the overall dining experience on campus.

1.2 Document Conventions

The document is written using the following terms.

Term	Description
User	KAIST members using the service authenticated via a school email
Manager	administrative staff updating menus and monitoring feedback

1.3 Intended Audience and Reading Suggestions

This document is intended for the development team to guide implementation and for project managers to oversee milestones. Section 2 provides a high-level overview for all readers, while Section 4 (System Features) and Section 5 (Nonfunctional Requirements) are critical for developers and QA testers to ensure the application logic and performance meet the required standards. Product managers are encouraged to distribute roles based on the functional specifications outlined here. We recommend that all readers start with the introduction to grasp the project's core concept before moving to the specific technical chapters that apply to their respective roles.

1.4 Product Scope

The software is specifically targeted at the KAIST community to provide a more interactive dining environment and options that facilitate meal selection. The system provides real-time access to cafeteria menus, prices, and availability. It includes a sophisticated feedback system that uses **receipt verification** to ensure review integrity. Additionally, it offers personalized filtering for allergies and dietary preferences, such as **Halal** options, and generates weekly meal recommendations based on aggregated user data.

1.5 References

CLOVA OCR API Documentation: https://api.ncloud-docs.com/docs/ai-services-ocrl

- Used for defining the Receipt Verification Logic and OCR service interface requirements.

2. Overall Description

2.1 Product Perspective

This service is a new, self-contained project consisting of a mobile-friendly frontend and a robust backend infrastructure, The backend utilizes a proper database to store dynamic meal status and user-generated feedback data, ensuring the system can handle a variety of data types efficiently. To ensure maximum accessibility, the frontend is designed as a responsive web application that adjusts to different screen sizes and resolutions on both mobile platforms (Android & iOS). The architecture is designed to be modular to allow for future integration with other KAIST campus services.

2.2 Product Functions

2.2.1 User Management

Users can register and log in using their KAIST email. Upon registration, users set their dietary profile, including allergy checklists and ingredient preferences, which the system uses to filter meal options.

2.2.2 Menu and Availability Tracking

The system displays daily and weekly menus categorized by cuisine type. Managers can update the availability of specific dishes in real-time, marking them as "Sold Out" when necessary.

2.2.3 Verified Review and Rating

The core function allows users to rate meals on a 5-star scale and provide detailed feedback. To maintain authenticity, a **receipt checking** mechanism is implemented, requiring users to verify their purchase before posting.

2.3 User Classes and Characteristics

1) Users (Students/University staff)**:**

Primary and general users who utilize the app to check menus, set allergy filters, and submit verified feedback..

2) Managers (Cafeteria Managers):

Responsible for updating menu data, marking availability, and monitoring user satisfaction trends.

2.4 Operating Environment

- Primarily Mobile device
- Website

2.5 Design and Implementation Constraints

The application shall operate as a responsive web app compatible with Chromium engine and Safari. It is optimized for mobile devices running Android 9+ or iOS 15+. NoSQL databases (e.g., MongoDB) to handle the flexible data structures are required for varying cafeteria menus and large volumes of user-generated reviews. The backend will be hosted on AWS to ensure high availability and scalability.

2.6 User Documentation

The system shall provide a comprehensive online user manual accessible within the application's "Help" section. This documentation will include step-by-step tutorials for new users A dedicated administrative guide will also be provided for cafeteria managers to facilitate the efficient use of the management dashboard. This guide will detail the procedures for real-time menu updates, the process for marking items as "Sold Out," and methods for interpreting the aggregated feedback data and ranking charts.

2.7 Assumptions and Dependencies

2.7.1 User-Side Assumption

It is assumed that all users have access to a mobile smartphone or a computing device equipped with a stable internet connection (WiFi or cellular data) to receive real-time updates. Furthermore, it is assumed that users possess a valid kiast.ac.kr email account, as this is the primary requirement for authentication and community exclusivity.

2.7.2 Infrastructure Dependencies

The system relies on the continuous availability of the KAIST email server to process registration and password reset requests. Also the operation depends heavily on external cloud infrastructure AWS, for hosting the backend server and the NoSQL database. Any service interruptions from the cloud provider will directly affect the availability of the application.

2.7.3 Technical Service Dependencies

The "Verified Review" feature is dependent on the performance and accuracy of a third-party Optical Character Recognition (OCR) API.

3. External Interface Requirements

3.1 User Interfaces

1) Login Page

The login page shall serve as the primary entry point for existing users, allowing them to access the system by entering their unique ID and password. This interface includes a "Forgot Password" link that initiates a recovery process via the user's registered KAIST email and provides a direct button to the registration page for new users who have not yet created an account.

2) Registration Page

The registration page is designed to facilitate the onboarding of new KAIST members. Users are required to input their desired ID, password, and a valid kaist.ac.kr email address, which must be verified through a link sent to their inbox. Additionally, this page confirms their agreement with the service's terms of use.

3) Dashboard Page

The Dashboard serves as the central hub for the application's meal discovery features. A map icon at the top right provides an immediate shortcut to the Map Page (3.1.4). A prominent search bar is centered in the upper section to allow for direct text-based meal inquiries. Below this, a horizontal tab-based navigation menu allows users to browse meals by specific cuisine categories. A dedicated filter icon is positioned on the right to allow for more granular dietary adjustments. Each card includes the trigger to navigate to the Meal Page (3.1.6).

3-1) Map Page

The map page provides a visual representation of the KAIST campus, marking the geographic locations of all operating cafeterias. Users can tap on specific markers to view a quick summary of the cafeteria's name and operating hours, helping them navigate to the nearest dining facility efficiently.

4) Search Page

The search page offers advanced filtering tools to help users find specific meals or cafeterias. It allows users to filter results by cuisine type (e.g., Korean, Western, Halal), specific cafeteria names.

5) Meal Page

The meal page provides an elaborate detail view of a specific dish selected from the dashboard or search results. It shall display the dish's price, a comprehensive list of ingredients, nutritional facts (calories, macronutrients), and clear allergy warnings. It also indicates the real-time "Availability" status to inform users if the meal is sold out.

5-1) Meal Review Page

This interface is dedicated to gathering user feedback and displaying existing evaluations. It includes a section for cumulative statistics, such as average ratings and the total number of reviews. Authenticated users can access an input form to provide a 1–5 rating, evaluate specific sub-items like taste and portion size, and write a short text review.

6) My Page

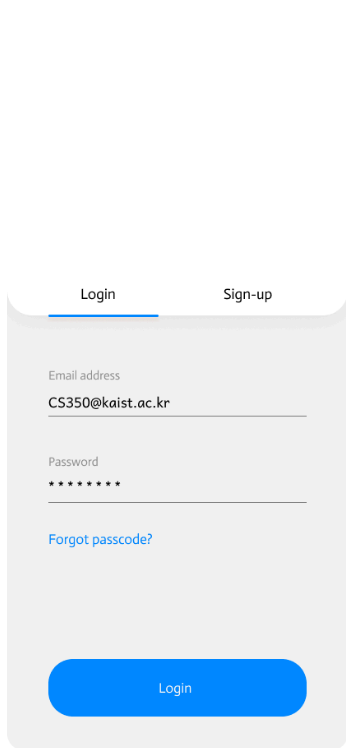
My Page allows users to manage their personal account information and view a summary of their activity. It acts as a central hub where users can view their current "Verified Review" count and access sub-pages.

7) My Review Page

This page serves as a personal archive for all feedback submitted by the user. Each entry displays the meal name, the date of the review, and the original text content. Most importantly, it features a response section for each feedback; when a cafeteria manager post a reply to the user's feedback, it is highlighted here with a distinct visual indicator, such as a "Replied" tag. This interface facilitates a two-way communication channel, allowing users to see how their suggestions are being addressed by the cafeteria staff.

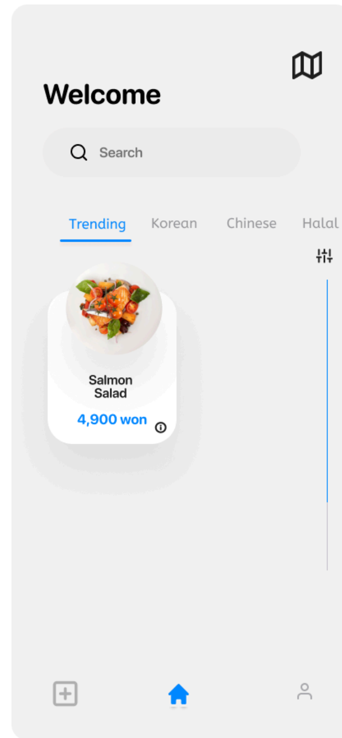
8) Receipt Page

The receipt page serves as the entry point for the "Verified Review" process. Users are prompted to upload a clear photo of their dining receipt to prove they have consumed the meal they wish to review. Once a receipt is uploaded, the verification page displays the results of the OCR scan for the user to confirm. The system identifies the date, cafeteria location, and the specific meal purchased. Upon confirmation of these details, the user is granted "Verified" status and redirected to the Meal Review Page to submit their authentic feedback.

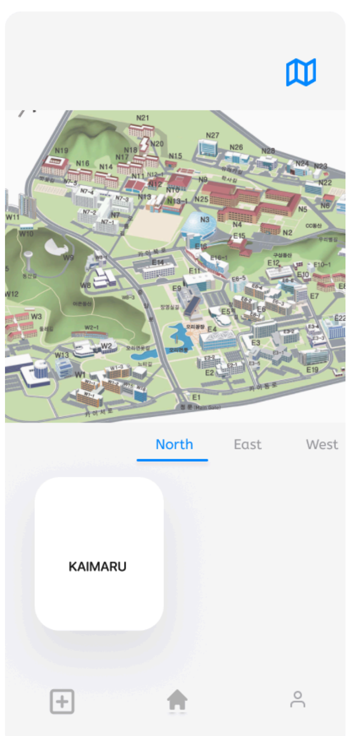


1) Login page

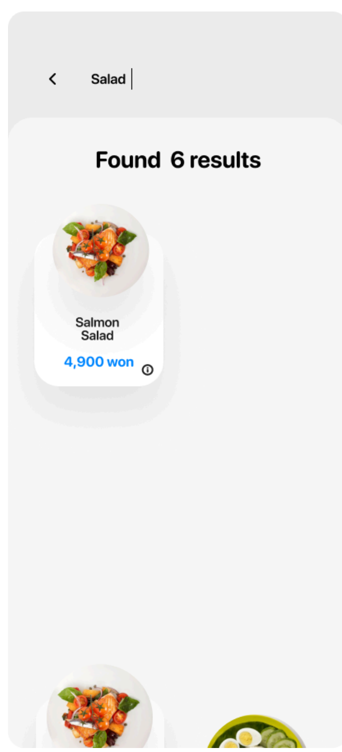
2) Registration page



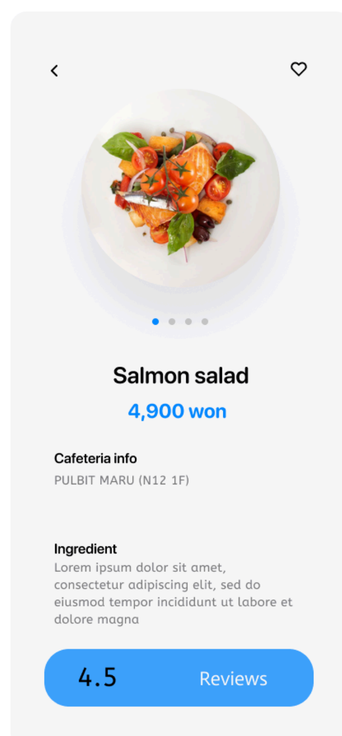
3)Dashboard page



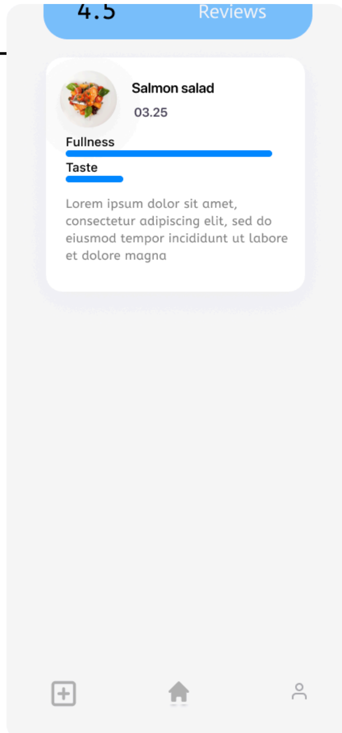
3-1) Map page



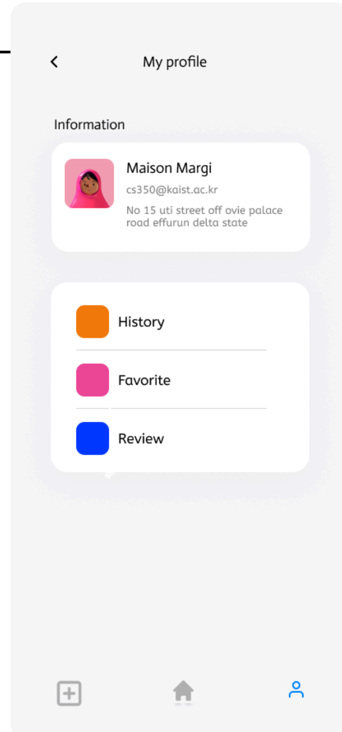
4) Search page



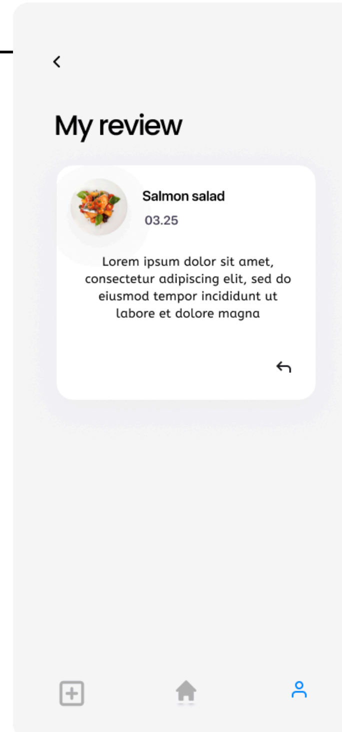
5) Meal page



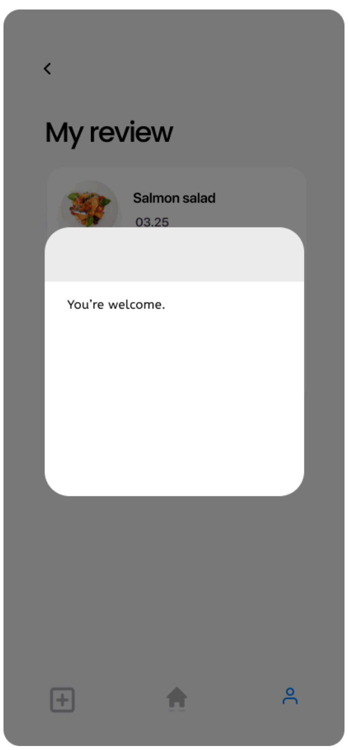
5-1) Meal review page



6) My page

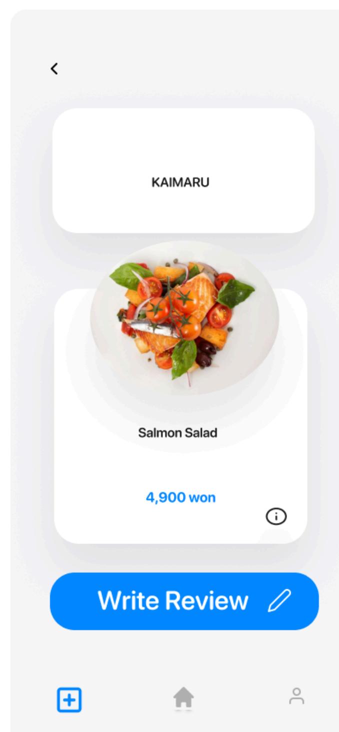


7) My review page



7-1) My page

8) Receipt page



8-1) Receipt page

3.2 Hardware Interfaces

The application is designed to interface with a wide range of modern smartphones. To ensure the receipt checking feature functions effectively, the device must be equipped with a high-resolution camera (minimum 8MP) capable of capturing clear images of printed receipts for OCR processing. The user interface is optimized for capacitive touchscreens with a minimum resolution of 720x1280 pixels to ensure that the Dashboard meal cards and Map markers are easily interactable.

The system's backend is hosted on cloud-based hardware, specifically AWS instances, to provide the computational power necessary for real-time menu filtering and feedback aggregation. The server hardware must be configured to handle the high throughput of concurrent connections expected during peak lunch and dinner hours on campus.

3.3 Software Interfaces

Again, the application shall interface with the Android (version 9.0+) and iOS (version 15.0+) operating systems. For the web-based components, the system must remain compatible with modern web engines, specifically Chrome (v106+) and Safari, to ensure that the React-based frontend renders correctly across different mobile platforms.

A critical software interface is the link to a third-party Optical Character Recognition (OCR) API (e.g., Google Cloud Vision or Naver Clova). The system shall send uploaded receipt images to this service and receive a structured text response. This interface is vital for the Receipt Verification Page (Section 3.1.8), as it allows the app to programmatically confirm that the user actually purchased the meal they are reviewing.

3.4 Communication Interfaces

The application interfaces with the KAIST Email Server via the SMTP protocol to send and verify registration links for the kaist.ac.kr domain. For intranet and internal communication between the web server and the database, the system follows the standard TCP/IP protocol suite. All API endpoints are designed to be stateless to improve scalability and ensure that the communication interface can handle the high-load scenarios defined in the Performance Requirements (Section 5.1).

4. System Features

4.1 User Authentication

4.1.1 Description and Priority

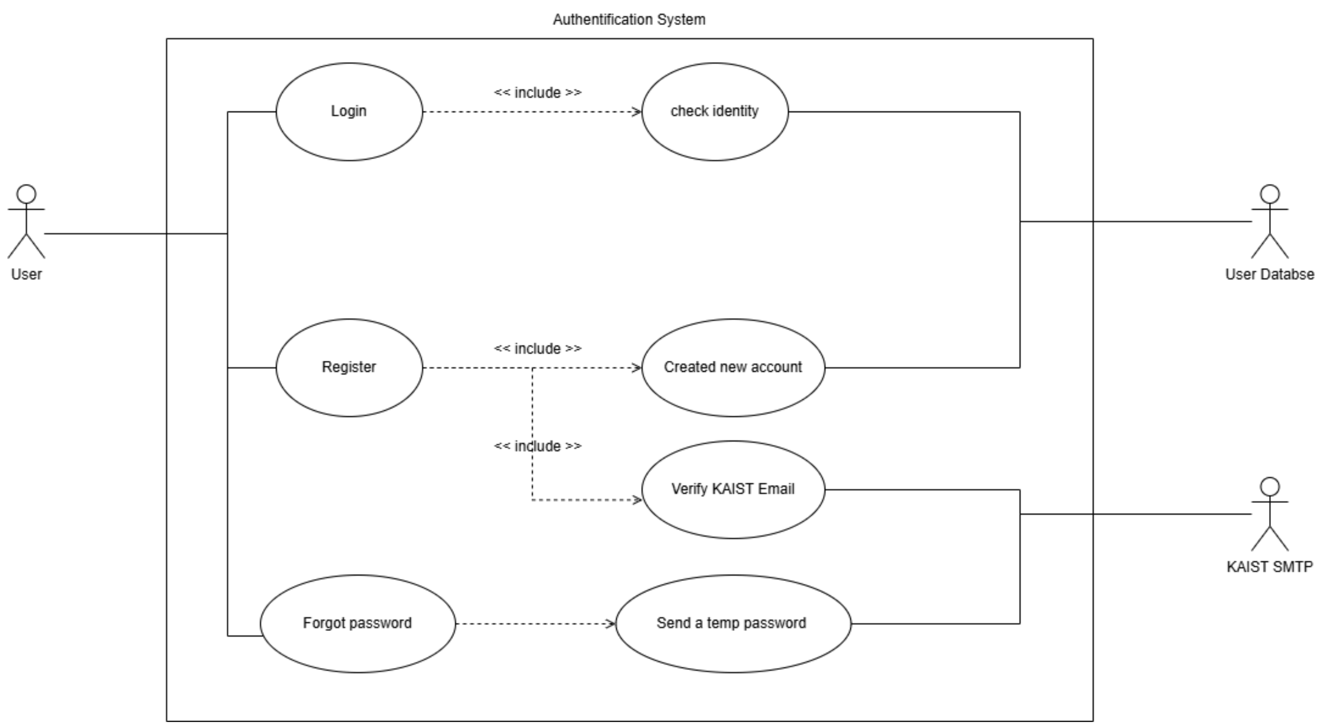
The User Authentication System serves as the secure gateway to the application, ensuring that only verified members of the KAIST community can access and contribute to the platform. This system manages account creation, secure login, and the verification of academic identity. Priority: High.

4.1.2 Stimulus/Response Sequences

1. Login: The user enters their unique ID and password; the system authenticates the credentials against the database and redirects the user to the Dashboard.
2. Registration: The user provides an ID, password, and KAIST email; the system sends a verification link to the email and activates the account upon confirmation.
3. Identity Verification: The system checks the email domain; if the domain is not @kaist.ac.kr the registration is blocked with an error message.

4.1.3 Functional Requirements

- REQ-1: The system shall provide a registration interface that collects user credentials and a mandatory KAIST email address.
- REQ-2: The system shall implement an automated email verification service to confirm the user's affiliation with the university.
- REQ-3: The system shall authenticate login requests and maintain session security using encrypted tokens to prevent unauthorized access.



4.2 Menu Information & Categorization System

4.2.1 Description and Priority

This feature provides comprehensive and categorized meal information across all campus locations. It is designed to help users plan their meals through detailed descriptions and real-time status updates. Priority: High.

4.2.2 Stimulus/Response Sequences

1. Browse Menu: The user selects a specific cuisine tab (e.g., Halal or Salad); the system retrieves and displays the relevant meal cards from the database.
2. View Details: The user taps the box on a meal card; the system displays the price, full ingredient list, and nutritional facts.
3. Availability Update: A cafeteria manager marks a dish as "Sold Out"; the system instantly updates the Dashboard status to inform all users.

4.2.3 Functional Requirements

- REQ-1: The system shall categorize all menus by cuisine type, cafeteria location, and specific dietary labels such as Halal or vegetarian options.
- REQ-2: The system shall display real-time availability status for every menu item, allowing managers to notify users of stock depletion immediately.

4.3 Verified Feedback & Response System

4.3.1 Description and Priority

The feedback system allows users to post authenticated reviews and enables cafeteria managers to engage with the community by providing official responses. This ensures transparency and reliability in meal evaluations. Priority: High.

4.3.2 Stimulus/Response Sequences

1. Verification: The user uploads a photo of their dining receipt; the system uses OCR to confirm the date and purchase details.
2. Feedback Posting: Upon successful verification, the user submits a rating; the system updates the average score of the meal and notifies the cafeteria manager.
3. Manager Response: The cafeteria manager replies to a review; the system updates the My Review Page and notifies the user.

4.3.3 Functional Requirements

- REQ-1: The system shall implement a Receipt Checking module using OCR technology to authenticate that a review is based on a real purchase.
- REQ-2: The system shall allow users to evaluate meals based on a 1-5 scale and itemized criteria including taste, price, and portion size.
- REQ-3: The system shall provide a dedicated interface for cafeteria managers to view verified feedback and post official responses.

4.4 Personalized Dietary Filter System

4.4.1 Description and Priority

This feature allows for a tailored user experience by filtering menus according to individual health requirements and culinary preferences. Priority: **Medium-High**.

4.4.2 Stimulus/Response Sequences

1. **Setup Profile:** The user completes the **Allergy Checklist** in their profile settings.
2. **Automated Filtering:** The user views the Dashboard; the system scans the ingredients of all meals and highlights or hides dishes that conflict with the user's allergies.
3. **Preference Toggle:** The user specifies a preference for specific ingredients; the system prioritizes meals containing these ingredients in the search results.

4.4.3 Functional Requirements

- **REQ-1:** The system shall allow users to manage a personal profile containing an exhaustive checklist of allergens and preferred/disliked ingredients.
- **REQ-2:** The system shall cross-reference the user's allergy data with the ingredients listed for each meal to generate automated safety warnings.

4.5 Analytics & Recommendation System

4.5.1 Description and Priority

The recommendation engine uses aggregated feedback data to guide users toward high-quality dining options and provides performance data to cafeteria managers. Priority: **Medium**.

4.5.2 Stimulus/Response Sequences

1. **Generate Ranking:** The system identifies the highest-rated meals from the past seven days.
2. **Display Recommendation:** The system features these meals in a "Weekly Recommendation" section on the Home Dashboard for all users.

4.5.3 Functional Requirements

- **REQ-1:** The system shall automatically calculate the "Weekly Best" meals based on the average star ratings and the volume of verified reviews.
- **REQ-2:** The system shall provide a visual ranking chart of cafeterias based on cumulative user satisfaction to help users choose the best dining locations.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system shall ensure that all page transitions and menu updates are completed within a maximum 2 seconds including client to server response time, server to DB, server background tasks processing time. The architecture must support up to 500 concurrent users during peak meal hours (11:30 AM to 1:00 PM).

5.2 Security Requirements

Registration is strictly limited to users with a kiast.ac.kr email address to ensure community exclusivity. All personal data, specifically allergy information scanned receipts and dining habits, must be encrypted during storage to protect user privacy. All network traffic shall be sent over HTTPS .

5.3 Software Quality Attributes

1) **Accuracy and Reliability**

The system must maintain a high level of accuracy, particularly regarding nutritional facts, calorie counts, and allergy warnings. The system shall ensure that the automated flagging of ingredients is 100% accurate based on the data provided by cafeteria managers. Furthermore, the reliability of the receipt checking mechanism must be high enough to correctly identify purchase details through OCR, preventing valid users from being incorrectly blocked from the feedback system.

2) **Usability and User-Friendliness**

As the application is designed for quick use during short breaks between classes, the user interface must be highly intuitive. The Dashboard Page (Section 3.1.3) shall be designed such that a user can find their preferred cuisine (e.g., **Salad** or **Halal**) and check the meal price within three taps. The use of recognizable icons, such as the map marker, is prioritized over complex text menus to ensure that international and domestic students can navigate the service with equal ease.

3) **Scalability and Availability**

The service shall be accessible 24/7 to allow students to check early breakfast menus or late-night operations. Because campus cafeterias experience extreme traffic spikes during the peak lunch hour (11:30AM – 1:00PM), the system architecture must be horizontally scalable. It shall maintain consistent performance levels even when more than 500 devices are simultaneously requesting real-time menu and congestion updates.

4) **Maintainability and Portability**

The codebase shall follow an Object-Oriented Programming (OOP) paradigm and be modularly structured to facilitate easy updates. This modularity is essential for adding new cafeterias or revising the "Cuisine Tab" categories as campus facilities are renovated. Additionally, the application shall remain portable across various mobile web environments, ensuring that updates to iOS or Android operating systems do not result in a loss of core functionality.

5.4 Business Rules

Only individuals with a valid and verified kaist.ac.kr email address are authorized to create an account and utilize the personalized features of the application. This rule ensures that the feedback and congestion data remain relevant to the actual campus residents and that the community remains secure from external influence.

To prevent the manipulation of cafeteria rankings through fake reviews, the system enforces a strict "One Receipt, One Review" policy. A user must pass the receipt check via the OCR verification page (Section 3.1.8) before the system grants them the authority to post a "Verified" review. Unverified users may browse menus but are restricted from contributing to the official average ratings.

The system identifies allergy information as sensitive personal health data. Under this business rule, the application shall never share an individual user's allergy profile with third parties or other users. The data is used exclusively for the internal logic to provide an automated safety menu on the user's device.